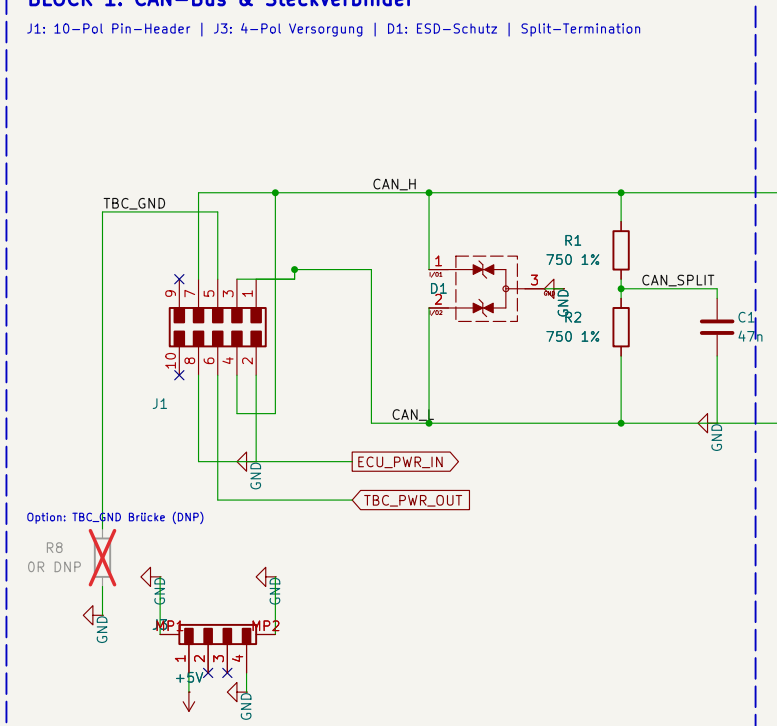
CAN-BUS SCHNITTSTELLE & SYSTEM-STECKVERBINDER

JC8012P4A1C – Blatt 1: CAN-Transceiver, Bus-Filterung und Display-Interface

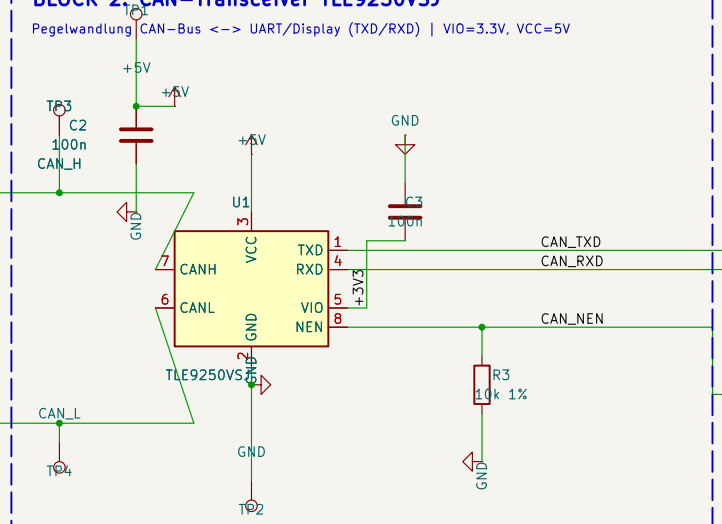
BLOCK 1: CAN-Bus & Steckverbinder

J1: 10-Pol Pin-Header | J3: 4-Pol Versorgung | D1: ESD-Schutz | Split-Termination



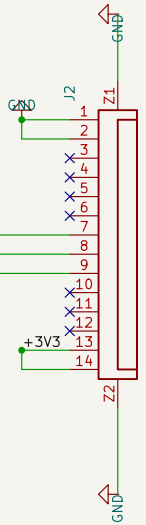
BLOCK 2: CAN-Transceiver TLE9250VSJ

Pegelwandlung CAN-Bus <-> UART/Display (TXD/RXD) | VIO=3.3V, VCC=5V



BLOCK 3: Display-Interface J2

J2: 14-Pol FPC Flachbandkabel zum JC8012 Display-Controller



HIERARCHISCHE UNTERSEITE: Stromversorgung

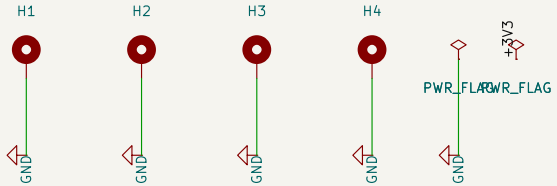
Buck 12V/24V -> 5V, Verpolungsschutz

Versorgung

Datei: power.kicad_sch

BEFESTIGUNG & TEST

H1-H4: Bohrungen M2 (Spacer L2.5) | FLG: ERC Power Flags



Sheet: /
File: JC8012P4A1C_I_W_Y_CAN.kicad_sch

Title:

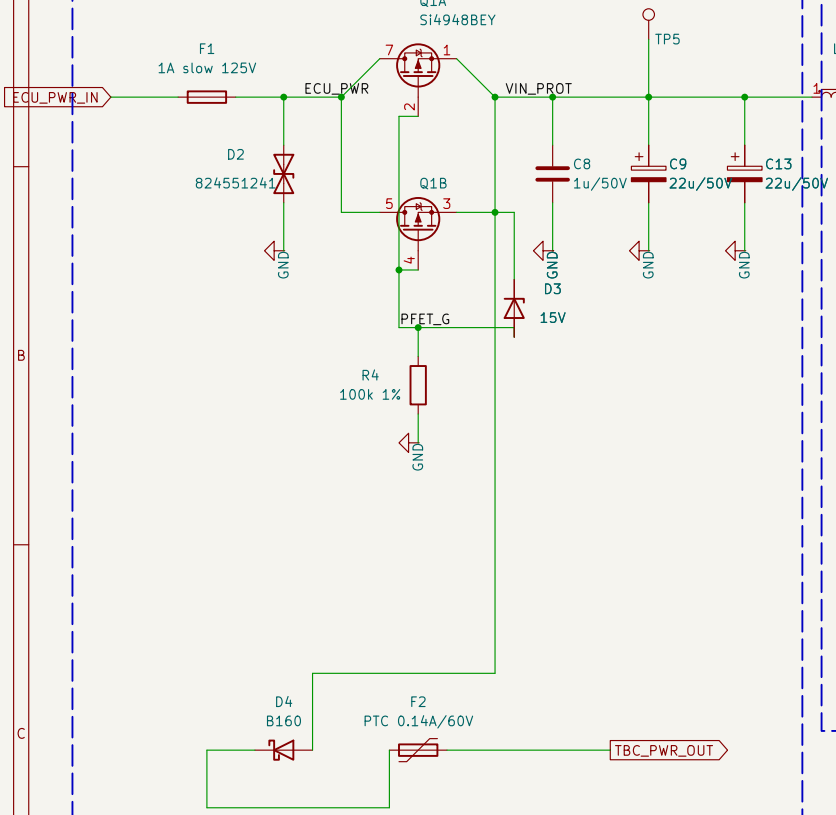
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 1/2

SPANNUNGSVERSORGUNG & SCHALTREGLER (BUCK)

JC8012P4A1C – Blatt 2: Verpolungsschutz, Eingangsfilterung und 5V MagI3C Buck-Regler

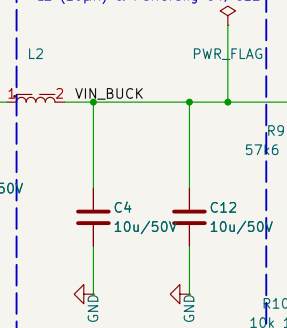
BLOCK 1: Eingangsschutz & Verpolungssperre

F1: 1A Sicherung | D2: TVS 30V | Q1: Dual P-FET Verpolungsschutz | D4+F2: TBC-Speisung



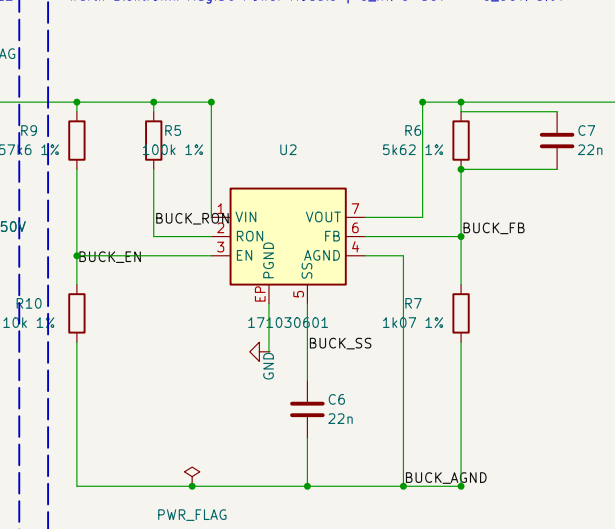
BLOCK 2: LC-Filter

L2 (10µH) & Pufferung C4, C12



BLOCK 3: MagI3C Buck-Regler 171030601

Würth Elektronik MagI3C Power Module | U_IN: 6–36V -> U_OUT: 5.0V



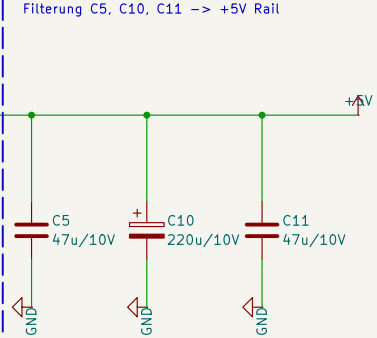
Layout-Hinweis BUCK_AGND:

R7, R10, C6 und Pin 4 (AGND) als ruhige analoge Sternmasse führen.

AGND ist intern im Würth-Modul mit PGND (Exposed Pad) verbunden.

BLOCK 4: 5V Ausgang

Filterung C5, C10, C11 -> +5V Rail



Sheet: /Versorgung/ File: power.kicad_sch		
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Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 2/2